Technology Stack

AI/ML and Generative AI Track • Project Documentation

Project	Trip Genius AI (VoyagePlan Travel Planner)	**Date**	26 October 2026
Submitted by	Kumkum Thakur (individual project)	**Hall Ticket No.**	0157CS241131
College	LNCTS	**University**	RGPV

Layer / purpose	Technology	Role
Frontend	React + Vite (JavaScript / JSX)	Renders pages, planner form, itinerary and My Trips
Styling	Project CSS / component styling	Responsive layout and visual presentation
Backend API	Python 3.11 + FastAPI	Validates requests and serves itinerary-generation endpoints
Validation / schemas	Pydantic	Typed request and response models
AI inference	Groq API	Generates itinerary content from trip preferences
Authentication	Supabase Auth	User sign-up, sign-in and session handling
Database	Supabase PostgreSQL	Stores saved trips (with ownership / RLS)
Frontend hosting	Vercel	Builds and serves the web application
Backend hosting	Render	Runs the FastAPI web service (uvicorn)
Version control	Git + GitHub	Source control: https://github.com/thakurkumkum/travel-itinerary-planner

Configuration (environment variables)

Variable	Where	Notes
VITE_API_BASE_URL	Vercel (frontend)	https://travel-itinerary-planner-1ysc.onrender.com (no /health or /api suffix)
VITE_SUPABASE_URL	Vercel (frontend)	Public Supabase project URL
VITE_SUPABASE_ANON_KEY	Vercel (frontend)	Public anon key only; protected by database policies
SUPABASE_URL / key variables	Render (backend)	Server-side Supabase access; service-role key must stay server-side
Groq API key variable	Render (backend)	Secret; never placed in a VITE_ variable or committed
ALLOWED_ORIGINS	Render (backend)	Exact Vercel origin for CORS

Security guidance

VITE_ variables are bundled into client-side code, so only public client configuration such as the Supabase anon key belongs there. Never place service-role keys or Groq secret keys in VITE_ variables and never commit .env files; use a .env.example with placeholders.